

3.1.2 Movement analysis

Students should develop knowledge and understanding of the basic principles of movement and their effect on performance in physical activity and sport.

3.1.2.1 Lever systems, examples of their use in activity and the mechanical advantage they provide in movement

Content	Additional information
First, second and third class lever systems within sporting examples	Identification of first, second and third class lever systems. Basic drawings of the three classes of lever to illustrate the positioning of: • fulcrum • load (resistance) • effort. Draw linear versions of a lever, showing the positioning of the fulcrum, load/resistance and effort. Students do not need to be taught to draw anatomical body parts but must be able to link the correct lever to a sporting movement or action. Interpretation of sporting movements or actions which involve flexion or extension of the elbow and/or knee, and plantar or dorsi-flexion at the ankle.

Content	Additional information
Mechanical advantage – an understanding of mechanical advantage in relation to the three lever systems	Label the effort arm and load/resistance arm on the three classes of lever. Mechanical advantage = effort arm ÷ weight (resistance) arm. Labelling of the effort arm and resistance arm on lever drawings, and interpretation of the mechanical advantage of that lever.
Analysis of basic movements in sporting examples	Types of movement: • flexion/extension at the shoulder, elbow, hip and knee • abduction/adduction at the shoulder • rotation of the shoulder • circumduction of the shoulder • plantar flexion/dorsiflexion at the ankle. This section links specific sporting actions to the types of movement. Applied anatomy and physiology (page 9) links the joint type to the type of movement only. This should include but not be limited to the following sporting actions: • elbow action in push-ups/football throw in • hip, knee and ankle action in running, kicking, standing vertical jump, basic squats • shoulder action during cricket bowling.

3.1.2.2 Planes and axes of movement

Content	Additional information
Identification of the relevant planes (frontal, transverse, sagittal) and axes (longitudinal, transverse, sagittal) of movement used whilst performing sporting actions	Planes (frontal, transverse, sagittal) and axes (longitudinal, transverse, sagittal) should be related to sporting actions. Teaching of these planes/axes should include but not be limited to the following sporting actions: • front somersault/forward roll/running action • 360° twist (ice skating spin)/discus thrower rotating in circle effort • cartwheel.